

PRODUCT REQUIREMENTS DOCUMENT

Navi Mumbai

House Price Prediction Platform

Empowering buyers, sellers & investors with AI-driven property valuations

Attribute	Details	Value
Version	1.0 — Initial Release	Feb 2026
Status	Draft — Pending Stakeholder Review	☐ Draft
Owner	Product Management	PM Team
Target Launch	Q3 2026 (Beta), Q4 2026 (GA)	Oct 2026
Geography	Navi Mumbai, Maharashtra, India	MH

1. Executive Summary

The Navi Mumbai House Price Prediction Platform is an AI/ML-powered web and mobile application that provides accurate, real-time property price estimates for residential properties across Navi Mumbai — including key nodes such as Kharghar, Nerul, Vashi, Belapur, Panvel, Airoli, Ghansoli, Kopar Khairane, and Turbhe.

The platform addresses a critical pain point in the Navi Mumbai real estate market: information asymmetry. Buyers overpay due to lack of data; sellers underprice due to opacity. By leveraging machine learning models trained on historical transaction data, CIDCO records, MahaRERA registrations, and live market listings, the platform delivers hyperlocal valuations with explainable confidence scores.

Key value propositions include instant property valuations, neighborhood trend analytics, investment ROI forecasts, and a comparison tool for side-by-side property analysis. The platform targets three primary user segments: homebuyers, real estate agents, and property investors.

2. Problem Statement

2.1 Market Context

Navi Mumbai is one of India's most planned satellite cities, with a growing population exceeding 1.1 million residents and a booming real estate sector driven by CIDCO developments, the upcoming Navi Mumbai International Airport (NMIA), and rapid infrastructure growth including metro expansion (Lines 1 & 2).

Despite strong demand, the residential property market suffers from a significant data gap:

- Transaction data is fragmented across IGR Maharashtra, MahaRERA, and private aggregators
- Listed prices on portals diverge significantly from actual registered prices
- Hyperlocal factors — node-level price differentials, proximity to upcoming metro stations, CIDCO plot status — are poorly understood by most buyers
- First-time buyers, NRI investors, and newcomers to the city lack access to professional valuation expertise

2.2 User Pain Points

User Segment	Core Problem	Current Workaround
Homebuyer	Cannot verify if asking price is fair	Relies on broker advice (biased)
Investor	No ROI forecasts for emerging nodes	Manual data scraping from portals
Real Estate Agent	Time-consuming CMA (Comparative Market Analysis)	Excel sheets / gut instinct
NRI Buyer	Remote purchase decisions with no trusted data source	Asks relatives; relies on expensive consultants

3. Goals & Success Metrics

3.1 Product Goals

- Deliver property valuations with a Mean Absolute Percentage Error (MAPE) below 8% across all Navi Mumbai nodes
- Reduce average user time-to-valuation from hours (manual research) to under 60 seconds
- Provide explainable AI outputs — users should understand the top 5 factors driving their estimate
- Achieve 70% weekly active user retention within 6 months of GA launch
- Generate 1,000+ verified agent subscriptions by end of Year 1

3.2 Key Performance Indicators

Metric	Definition	Target (6M)	Target (12M)
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Model MAPE	Mean Absolute % Error	< 10%	< 8%
Monthly Active Users	Unique users/month	10,000	50,000
Valuations Generated	Total estimates generated	25,000/month	150,000/month
Agent Subscriptions	Paid Pro plan signups	300	1,000
NPS Score	Net Promoter Score	> 40	> 55
Data Coverage	% of Navi Mumbai nodes covered	80%	100%

4. User Personas

Persona 1 — Priya, First-Time Homebuyer

Age / Location	32 years old, currently renting in Kharghar
Goal	Buy a 2BHK flat in Kharghar or Nerul within ₹75–90 lakh budget
Frustration	Cannot tell if a ₹85L listing is overpriced; broker says it's a 'fair deal'
Needs	Instant price check, fair value certificate, nearby comparable sales

Persona 2 — Rajan, Real Estate Agent

Age / Location	44 years old, 15+ years experience, operates in Vashi and Belapur
Goal	Close deals faster with data-backed valuations; build client trust
Frustration	Spends 2–3 hours per CMA; no professional tool for Navi Mumbai micro-markets
Needs	Bulk valuation tool, branded PDF reports, API access for CRM integration

Persona 3 — Deepak, NRI Investor

Age / Location	48 years old, based in Dubai; family in Navi Mumbai
Goal	Invest ₹1.5–2 Cr in a Panvel or Kharghar property for long-term appreciation

Frustration	Cannot physically visit; fears overpaying or buying in wrong node given NMIA impact
Needs	5-year price forecasts, node comparison, proximity-to-NMIA valuation impact analysis

5. Scope & Features

5.1 In-Scope (MVP — Q3 2026)

- Instant property valuation engine (address/pincode input → price per sq ft + total estimate)
- Coverage of all 9 major Navi Mumbai nodes: Vashi, Nerul, Belapur, Kharghar, Panvel, Airoli, Ghansoli, Kopar Khairane, Turbhe
- Property type support: Apartments (1BHK, 2BHK, 3BHK, 4BHK), Row Houses, CIDCO plots
- Key input parameters: area (sq ft/sq m), floor, building age, parking, furnished/unfurnished status, facing direction
- Confidence score (High / Medium / Low) with explanation of top 5 value drivers
- Comparable sales: 5 most similar recent transactions within 1km radius
- Node-level price trend charts (3M, 6M, 1Y, 3Y timeframes)
- Free tier: 5 valuations/month; Pro tier (₹499/month): unlimited valuations + PDF export
- Mobile-responsive web application (PWA); iOS and Android apps Phase 2

5.2 Phase 2 Features (Q4 2026 — Q1 2027)

- 5-year price appreciation forecast by node (investment ROI calculator)
- NMIA proximity impact analysis — radius-based price uplift estimates
- Metro Line 1 & 2 station proximity scoring
- Agent Pro dashboard: bulk valuations, CRM webhooks, white-label PDF reports
- Portfolio tracker for investors: monitor value of owned properties over time
- API access (REST) for enterprise clients and prop-tech integrations
- Vernacular language support: Marathi and Hindi interfaces

5.3 Out of Scope

- Commercial or industrial property valuations
- Greater Mumbai (BMC area) or Thane District (except Navi Mumbai nodes)
- Legal due diligence, title verification, or loan advisory services
- Real estate transaction / brokerage marketplace

6. Data Sources & ML Architecture

6.1 Data Sources

Source	Data Type	Refresh Frequency
IGR Maharashtra	Registered sale deed prices (last 5 years)	Monthly
MahaRERA	Project registrations, unit inventories, completion status	Weekly
CIDCO	Plot allotments, scheme prices, node master plan data	Quarterly
Portal Listings (MagicBricks, 99acres, Housing.com)	Listed prices, property specs, amenities	Daily
OpenStreetMap / Google Maps API	Distance to metro, schools, malls, hospitals, NMIA	Monthly
NMMC / CIDCO Node Plans	Infrastructure development timelines	Quarterly

6.2 ML Model Strategy

The prediction engine will use an ensemble approach combining:

- Gradient Boosted Trees (XGBoost / LightGBM) as the primary regression model — handles mixed feature types and non-linear pricing patterns well for Indian real estate
- Geospatial features: Distance to metro stations, NMIA, node center, coastline, CIDCO sector boundaries encoded as continuous features
- Time-series trend component: SARIMA-based node-level price index, combined with property-specific features for temporal adjustment
- Explainability layer: SHAP (SHapley Additive exPlanations) values surfaced to end users as "Top 5 factors affecting your property's value"

Model training will use a rolling 5-year window of IGR-registered transactions (approximately 40,000–60,000 valid records for Navi Mumbai). Re-training will occur monthly with new transaction data to capture market shifts.

7. Functional Requirements

7.1 Valuation Engine

ID	Requirement	Priority	Acceptance Criteria
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VE-01	System shall accept property address or pincode as primary location input	Must Have	Address resolves to GPS coordinates within 50m accuracy
VE-02	System shall accept carpet area in sq ft (with sq m conversion)	Must Have	Area input validates 100–10,000 sq ft range
VE-03	System shall generate price estimate in ₹ per sq ft and total ₹ within 5 seconds	Must Have	P95 latency ≤ 5s for 95% of requests
VE-04	System shall display confidence band ($\pm X\%$) alongside point estimate	Must Have	Confidence interval shown for all outputs
VE-05	System shall surface top 5 SHAP-driven value factors in plain English	Must Have	Factors shown with directional arrows ($\uparrow$ premium / $\downarrow$ discount)
VE-06	System shall show 5 comparable transactions within 1km radius and 12 months	Should Have	Comps displayed with address, area, price per sq ft, date
VE-07	Users shall be able to save valuations to their profile history	Should Have	Save button stores valuation; accessible in Dashboard

7.2 Market Analytics

- Node-level price per sq ft trend charts for 3M / 6M / 1Y / 3Y timeframes
- Heatmap of Navi Mumbai showing price intensity by node and sector
- Year-over-Year and Month-over-Month appreciation rates per node
- Inventory supply indicator: New project launches vs absorption rate by node

7.3 User Accounts & Plans

- Free Tier: Google / Phone OTP sign-in, 5 valuations/month, basic comparables
- Pro Tier (₹499/month or ₹4,999/year): Unlimited valuations, PDF export, portfolio tracker, price alerts, node watchlist
- Agent Pro (₹1,499/month): All Pro features + bulk valuation (up to 50/day), branded PDF reports, CRM webhook integration (Phase 2)

8. Non-Functional Requirements

Category	Requirement	Target
Performance	Valuation API response time	≤ 3s p50, ≤ 5s p95

Availability	Platform uptime SLA	99.5% monthly
Scalability	Concurrent users during peak (post-NMIA news events)	5,000 concurrent users
Model Accuracy	Mean Absolute Percentage Error on holdout set	< 8% MAPE
Data Privacy	No PII stored against property addresses; PDPB 2023 compliant	Legal + Security review
Security	HTTPS everywhere, OAuth 2.0, rate limiting on valuation API	Penetration test before GA
Accessibility	WCAG 2.1 AA compliance	Audit before launch
Browser Support	Chrome, Safari, Firefox, Edge (latest 2 major versions)	Cross-browser QA

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
IGR data gaps / delayed updates create stale training data	Medium	High	Supplement with portal listing data; display data freshness warning
Model performs poorly in low-transaction nodes (e.g., Turbhe)	Medium	Medium	Fallback to node-level average + manual override flag; low confidence label
Legal challenges: displaying registered prices without IGR consent	Low	High	Engage legal counsel; use aggregated / anonymized data only
NMIA or infrastructure delays alter forecast validity	Medium	Medium	Monthly model re-training; scenario-based forecast ranges
Competition from larger portals (MagicBricks, NoBroker) launching valuation tools	High	Medium	Differentiate on hyperlocal accuracy, explainability, and

User trust gap: ML estimates seen as unreliable vs. broker advice	High	High	Navi Mumbai specialization Backtesting dashboard showing historical accuracy; user testimonials; confidence bands
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10. Delivery Roadmap

Phase	Timeline	Deliverables	Milestone
Phase 0 — Data & Foundation	Mar – Apr 2026	Data pipeline setup, IGR/MahaRERA ingestion, EDA & baseline model	Data Readiness ✓
Phase 1 — MVP Build	May – Jul 2026	ML model v1, core valuation API, web frontend (PWA), Free & Pro plans	Private Beta Launch
Phase 2 — Beta & Iteration	Aug – Sep 2026	Beta user feedback, model tuning, comparables feature, mobile optimization	Open Beta (1,000 users)
Phase 3 — GA Launch	Oct 2026	Public launch, marketing campaign, Agent Pro plan, PR & media coverage	General Availability ✓
Phase 4 — Growth	Nov 2026 – Q1 2027	Forecast engine, NMIA impact scores, API access, iOS & Android apps	50,000 MAU target

11. Stakeholders & Approvals

Stakeholder	Role	Responsibility	Status
Product Manager	Document Owner	PRD ownership, roadmap, prioritization	Approved
Head of Data Science	ML Lead	Model architecture, accuracy targets, data pipeline	Pending ☐

Engineering Lead	Tech Lead	System design, API, infra, scalability	Pending ☐
Legal & Compliance	Advisor	Data licensing, IGR usage rights, PDPB compliance	In Review
Growth & Marketing	Go-To-Market	Launch strategy, agent partnerships, user acquisition	Pending ☐

12. Appendix

A. Navi Mumbai Node Reference

Coverage priority order for model training data (based on transaction volume):

- Tier 1 (High Volume): Kharghar, Nerul, Vashi, Belapur, Panvel
- Tier 2 (Medium Volume): Airoli, Kopar Khairane, Ghansoli
- Tier 3 (Lower Volume): Turbhe, Sanpada, Ulwe (Phase 2 expansion)

B. Glossary

- MAPE — Mean Absolute Percentage Error: average percentage difference between predicted and actual prices
- SHAP — SHapley Additive exPlanations: ML interpretability framework attributing prediction to individual features
- CIDCO — City and Industrial Development Corporation: planning authority for Navi Mumbai
- MahaRERA — Maharashtra Real Estate Regulatory Authority: regulates real estate projects under RERA
- IGR — Inspector General of Registration: Maharashtra government body recording property transactions
- NMIA — Navi Mumbai International Airport: upcoming international airport expected to drive significant price appreciation in Panvel and Ulwe nodes
- CMA — Comparative Market Analysis: method of estimating property value using recent comparable sales
- PWA — Progressive Web App: web application with native app-like features including offline capability

C. Assumptions

- IGR Maharashtra will continue to make transaction data accessible (either publicly or via licensed API)
- MahaRERA data can be scraped or licensed for model training purposes
- NMIA Phase 1 operations commence by 2028 as per current government timelines

- Market conditions do not undergo a black swan event (major economic crisis, regulatory freeze) during the development period

End of Document — Navi Mumbai House Price Prediction PRD v1.0